

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 | SUB-STRAND 2.1: SIMILARITY AND ENLARGEMENT

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change when we do so? Achieng is a Grade 10 student in Kisumu. She needs passport photos for KCSE registration and a school ID. Mr Otieno, the photographer, prints her photo as stamp size S (3 cm x 4 cm, KES 30), postcard size P (6 cm x 8 cm, KES 120) and large size L (9 cm x 12 cm). Mr Otieno charges KES 2.50 per cm² of paper. Use the data and the ideas from our 12 lessons to answer each question in the space under it. Show your working, write units (cm, cm², cm³, KES, L), and use the words object, image, similar, LSF, ASF and VSF correctly. Write in full sentences when you are asked to explain.

Part 1: Same shape, different size

Print S is 3 cm x 4 cm with a diagonal of 5 cm. Print P is 6 cm x 8 cm with a diagonal of 10 cm. Mr Otieno also made a photo that is 6 cm x 4 cm by doubling only the width of S. (a) Using numbers, test whether P is similar to S. (b) Test whether the 6 cm x 4 cm photo is similar to S. (c) Explain in your own words what 'similar' means, and why Achieng's photo can be printed in different sizes without looking stretched.

(a) Both S and P are rectangles, so all corresponding angles are 90 degrees and equal. The sides are in the same ratio: width $6/3 = 2$, height $8/4 = 2$, diagonal $10/5 = 2$. Angles are equal and all sides have the same ratio, so P is similar to S. (b) The 6 cm x 4 cm photo also has four right angles, but the width ratio is $6/3 = 2$ and the height ratio is $4/4 = 1$. The ratios are different, so it is not similar to S. It is stretched. (c) Similar figures have the same shape but the size may differ. Corresponding angles must be equal AND corresponding sides must be in the same ratio. Achieng's photo can be printed bigger or smaller because every length is multiplied by the same number, so no direction is stretched more than another. (Congruent would mean the same shape and the same size.)

Part 2: The linear scale factor

(a) Treat S as the object and P as the image. Write the formula for the linear scale factor k and use it to find k for S to P using the width, the height and the diagonal. (b) Find k for S to L. (c) Mr Otieno wants to turn P

(a) $k = \text{image length} / \text{object length}$, using the same units and corresponding sides in the same order. Width: $6/3 = 2$. Height: $8/4 = 2$. Diagonal: $10/5 = 2$. So $k = 2$ for S to P. (b) S to L: $9/3 = 3$ and $12/4 = 3$, so $k = 3$. (c) P to S: $3/6 = 1/2$ (also $4/8 = 1/2$ and $5/10 = 1/2$). Since $0 < k < 1$ this is a reduction, which is still an enlargement with a fractional scale factor. (d) She is not right. Adding 3 cm to each side of S gives 6 cm x 7 cm, not 6 cm x 8 cm. The ratios would be $6/3 = 2$ and $7/4 = 1.75$, which are not the same, so the photo would be stretched and not similar to S. Enlargement works by multiplying every length by the same k, not by adding. Here every length is multiplied by 2: 3 to 6, 4 to 8 and diagonal 5 to 10. Adding 3 cm to the width and 4 cm to the height happens to give 6 x 8 only because 3 and 4 are exactly the amounts that doubling adds. It would not work for other sizes, such as L, which needs 6

back into S. Find k for P to S and say whether this is an enlargement or a reduction. (d) Achieng's friend says, 'Mr Otieno made P by adding 3 cm to each side of S.' Is she right? Explain using the lengths of S and P.	cm added to the width and 8 cm to the height.
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Part 3: Area and cost	
(a) Find the area of S, P and L in cm^2 . (b) Find the cost of each print at KES 2.50 per cm^2 . (c) Compare P with S. By how many times are the lengths multiplied, the area multiplied and the cost multiplied? (d) Explain why the price of P is KES 120 and not KES 60. (e) Without measuring, use $k = 3$ to predict the cost of L compared with S and check it with your answer to (b).	(a) S: $3 \times 4 = 12 \text{ cm}^2$. P: $6 \times 8 = 48 \text{ cm}^2$. L: $9 \times 12 = 108 \text{ cm}^2$. (b) S: $12 \times 2.50 = \text{KES } 30$. P: $48 \times 2.50 = \text{KES } 120$. L: $108 \times 2.50 = \text{KES } 270$. (c) From S to P the lengths are multiplied by 2 ($k = 2$). The area is multiplied by $48/12 = 4$, and the cost by $120/30 = 4$. (d) The ASF = $k^2 = 2^2 = 4$. Doubling the length and the width makes the picture 4 times as big in area, and Mr Otieno charges for the area of paper, not for the length. So the price is $4 \times \text{KES } 30 = \text{KES } 120$, not $2 \times \text{KES } 30 = \text{KES } 60$. (e) For $k = 3$, ASF = $3^2 = 9$, and cost follows area, so L should cost $9 \times \text{KES } 30 = \text{KES } 270$. This matches $108 \times 2.50 = \text{KES } 270$.

Part 4: Beyond photos	
(a) A small jerrycan holds 5 L. A similar larger jerrycan has every length twice as big (LSF 2). How much does it hold? Show how you worked it out. (b) A cube has side 2 cm. Find the volume of a similar cube with LSF 3. (c) On a map with scale 1:5,000,000, the road from Nairobi to Mombasa measures 8.8 cm. Find the real distance in km. (d) A classroom plan is drawn to scale 1:50 and measures 16 cm x 12 cm. The real room is 8 m x 6 m. Floor tiles cost KES 800 per m^2 . Find the cost of tiling the floor. (e)	(a) VSF = $k^3 = 2^3 = 8$. Volume = $5 \times 8 = 40 \text{ L}$. The length, width and height each double, so the volume is multiplied by $2 \times 2 \times 2 = 8$. (b) The 2 cm cube has volume $2 \times 2 \times 2 = 8 \text{ cm}^3$. With $k = 3$ the side is 6 cm and the volume is $6 \times 6 \times 6 = 216 \text{ cm}^3$, which is $27 = 3^3$ times as much. (c) Real distance = map length $\times n = 8.8 \times 5,000,000 = 44,000,000 \text{ cm} = 440 \text{ km}$. (d) The real floor area is $8 \times 6 = 48 \text{ m}^2$. Check with ASF: $192 \text{ cm}^2 \times 50^2 = 192 \times 2500 = 480,000 \text{ cm}^2 = 48 \text{ m}^2$. Cost = $48 \times \text{KES } 800 = \text{KES } 38,400$. (e) The length is multiplied by k , the area by k^2 (ASF) and the volume by k^3 (VSF). This works only for similar figures or objects, where every length is multiplied by the same k .

Copy and complete: when every length is multiplied by k , the length is multiplied by ____, the area by ____ and the volume by _____. State when this rule works.	
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Part 5: Putting the whole explanation together

Answer the driving question: How can a passport photo be made bigger or smaller without changing its shape, and how do its length, area and cost change when we do so? Write a clear explanation for a classmate who missed the lessons. Use Achieng's prints S, P and L, the 6 cm x 4 cm photo, and the jerrycan or another example. Include (a) the rule for keeping the shape, (b) how length, area and cost change, with numbers, and (c) what goes wrong if only the width is doubled.	A passport photo can be made bigger or smaller without changing its shape by multiplying every length by the same linear scale factor k, where $k = \text{image length} / \text{object length}$. This is an enlargement. It needs a centre and a scale factor, and then the corresponding angles stay equal and the corresponding sides are in the same ratio, so the image is similar to the object. For Achieng's photo, S to P has $k = 2$: 3 cm to 6 cm, 4 cm to 8 cm and the diagonal 5 cm to 10 cm, with all angles staying 90 degrees. Lengths change by k. The area changes by the ASF = $k^2 = 4$, so it goes from 12 cm² to 48 cm². Mr Otieno charges KES 2.50 per cm², so the price goes from KES 30 to KES 120. This is four times, not double, because cost follows area and not length. For S to L, $k = 3$, so the ASF = 9: the area is 108 cm² and the price is KES 270. Going from P to S uses $k = 1/2$, which halves the lengths. If only the width is doubled, the 6 cm x 4 cm photo has a width ratio of 2 but a height ratio of 1. The ratios are different, so the photo is stretched and not similar, and the k^2 rule does not apply: its area is only 24 cm², twice that of S. The same pattern works for solids: with LSF 2 the volume is multiplied by $k^3 = 8$, so a 5 L jerrycan becomes a 40 L jerrycan.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Testing and explaining similarity	Checks both equal angles and the same ratio of all corresponding sides with correct numbers, and correctly shows the 6 x 4 photo fails because the ratios are 2 and 1. Explains similar versus congruent accurately.	Checks angles and ratios with mostly correct numbers and identifies the distorted photo as not similar, but the explanation of similar is incomplete.	Uses only looks or one ratio to decide, or does not identify why the 6 x 4 photo is not similar.
Linear scale factor calculations	Uses $k = \text{image/object}$ with corresponding sides and the same units to get $k = 2, 3$ and $1/2$ correctly, classifies the reduction, and explains that multiplying (not adding) keeps the shape.	Calculates most scale factors correctly but gives a weak explanation of multiplying versus adding or of the reduction.	Confuses object and image, inverts the ratio, or uses addition to find the new lengths.
Area, volume and cost reasoning	Correctly finds the areas 12, 48 and 108 cm ² and the costs KES 30, 120 and 270, and links ASF = k^2 and VSF = k^3 to	Finds the areas and costs correctly and states k^2 and k^3 , but the link between cost and area or the volume reasoning is	Says the cost or area simply doubles, or does not use k^2 or k^3 .

	the cost and the jerrycan, explaining that cost follows area.	thin.	
Applying scale to real contexts	Converts the map scale and the scale drawing correctly (440 km, 48 m ² , KES 38,400) with correct units and shows the use of n for length and n^2 for area.	Gets most conversions correct but makes a unit or working slip, or does not show the use of n^2 .	Does not convert units correctly or does not apply the scale to length and area.
Final explanation and scientific language	Gives a connected, complete answer to the driving question using object, image, similar, LSF, ASF, VSF and Achieng's data, and addresses the stretched photo.	Answers the driving question with mostly correct terms and data but misses one link, for example the stretched photo or the cost.	Gives a list of facts without connecting length, area and cost or without using the key terms.

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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